

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING

COURSE CODE: CSA4305

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COURSE OUTCOME COVERED

CO-2 : Apply CSS layout and styling techniques to create visually appealing and responsive web interfaces, and use JavaScript to enable interactive client-side behavior. (L3)

Assessment Tool : Short Answer Test

Weightage: 10%

QUESTIONS:

Total Marks: 20

Question 1 (5 Marks)

Suppose that a company wants its website to work seamlessly on mobile, tablet, and desktop devices. Explain the concept of responsive web design and its importance. Describe the key CSS techniques used to achieve responsiveness, and explain how CSS Flexbox or Grid can be used to design adaptive layouts. Illustrate your answer with a simple example layout and suggest improvements to enhance user experience across different devices.

Question 2 (5 Marks)

Suppose that a webpage contains a form where user input must be validated before submission. Explain the concept of client-side validation and its purpose. Describe how JavaScript event handling is used in validation and explain the role of regular expressions in validating inputs such as email or phone numbers. Provide a suitable example and suggest improvements for better usability and security.

Question 3 (5 Marks)

Suppose that a webpage layout breaks when the browser window is resized. Explain the CSS box model and its components, and describe how CSS positioning techniques affect layout structure. Identify possible causes of layout breaking and explain how media queries can be used to resolve these issues. Suggest best practices to ensure a stable and responsive layout.

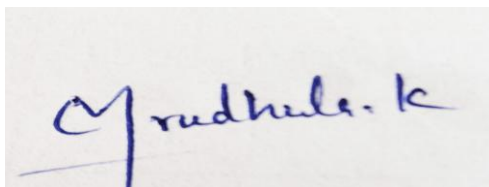
Question 4 (5 Marks)

Suppose that a website requires dynamic content updates without reloading the entire page. Explain the concept of the Document Object Model (DOM) and its role in web development. Describe how JavaScript can be used to manipulate the DOM for updating webpage content dynamically. Provide an example and suggest improvements to enhance performance and user experience.

Code of Conduct:

I K Mrudhulla(192372290) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

A photograph of a handwritten signature in blue ink on a white background. The signature appears to be 'Mrudhulla. K' with a stylized initial 'M'.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	10	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

CO₂-at₃

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Course code:- CSA4305 - Internet programming

Responsive web design (RWD) :-

Responsive web design (RWD) is a web development approach that ensures a website automatically adapts its layout, images and content to different screen sizes and devices such as desktops, laptops, tablets and smartphones. The main goal is to provide users with a consistent, attractive, and user-friendly experience without creating separate websites for each device.

Importance of Responsive web design:-

- Provides an optimal viewing experience on all devices
- Improves usability and accessibility
- Reduces development and maintenance costs by maintaining a single website
- Improves search engine ranking because search engines prefer mobile-friendly websites
- Increases user engagement and customer satisfaction
- Ensures faster loading and better overall performance.

Key CSS Techniques for Responsiveness:-

1. media queries:

- apply different CSS rules based on screen size.

CSS:

@ media (max-width : 768 px) {

• container {

flex-direction : column;

2. flexible layouts:-

- use percentage (%) widths instead of fixed pixels.

3. flexible images:-

img {

max-width: 100%;

height: auto;

}

4. Relative units:-

- use %, rem, em, and vm/vh instead of fixed px values

5. viewport meta tag:-

<meta name="viewport"

content="width=device-width,

initial-scale=1.0">

Using CSS flexbox for adaptive layouts:-

Flexbox allows elements to automatically adjust their size and position according to the available screen space

Example:-

```
<div class="container">
```

```
<div class="box">Home </div>
```

```
<div class="box">products </div>
```

```
<div class="box">contact </div>
```

```
</div>
```

• container {

display: flex;

justify-content: space-around;

flex-wrap: wrap;

• box {

width: 30%;

padding: 20px;

```

background : light blue ;
margin : 10px ;
}
@media (max-width : 768px) {
    .box {
        width : 100% ;
    }
}

```

Working:-

- on desktop, the boxes appear side by side
- on tablets, they adjust automatically.
- on mobile devices, each box occupies the full width.

Improvements for better user Experience:-

- use responsive navigation menus (hamburger menu on mobiles)
- optimize images and videos to reduce loading time
- use readable font sizes and proper spacing.
- provide touch-friendly buttons with sufficient size.
- Test the website on multiple devices and browsers
- Improve accessibility by using proper color contrast and Semantic HTML.

Conclusion:-

Responsive web design ensures that websites provide an excellent viewing experience across all devices. By using CSS techniques such as media queries, flexible layouts, responsive images, and flexbox or grid, developers can build adaptive websites that are visually appealing, easy to use, and perform efficiently on desktops, tablets and smartphones.

② client - side validation:-

client - side validation is the process of checking user input in the web browser before the form is submitted to the Server. it is mainly performed using Javascript and HTML validation attributes. The purpose is to ensure that users enter valid and complete information, reducing unnecessary Server requests and improving the overall user experience.

Purpose of client-side validation:-

- prevents invalid or incomplete data from being submitted
- provides instant feedback to users
- Reduces Server load by filtering incorrect inputs
- Improves website performance and user experience
- minimizes common input errors.

JavaScript Event handling in validation:-

JavaScript uses event handler to validate form fields during user interaction

- onsubmit:- validates the form before submission
- oninput:- validates data as the user types
- onblur:- checks input when the user leaves a field
- onclick:- Executes validation when a button is clicked.

Example:-

```
<form onsubmit="return validateform()">
  Email:
  <input type="text" id="email">
  <input type="Submit" value="submit">
</form>
<script>
  function validateform() {
```

```

let
email = document.getElementById("email").value;
if (email == " ") {
    alert("email cannot be empty");
    return false;
}
return true;
}
</script>

```

Role of Regular Expressions [Regex]:

Regular Expressions (Regex) are patterns used to verify whether input follows a required format.

Email Validation:-

```

let
email pattern = /^[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+
+\. [A-Za-z]{2,}$|/;

```

Phone Number Validation:-

```

let phone pattern = /^[0-9]{10}$|/;

```

These expressions ensure that:

- The email follows a valid email format
- The phone number contains exactly 10 digits

Improvements for better usability and Security:-

- Display clear error messages near the input fields
- Highlight invalid fields with colors or icons
- use HTML5 validation attributes such as required, type="email", and pattern
- Sanitize user input to prevent malicious data
- use HTTPS to protect sensitive data/information during

transmission.

Conclusion:-

client-side validation improves the quality of user input by checking data before it is sent to the server. Java script event handling enables real-time validation, while regular expressions verify formats such as email addresses and phone numbers. Combining client-side and server-side validation results in secure, efficient and user-friendly web applications.

Q3:- CSS Box model:-

The CSS Box model is the fundamental concept that defines how every HTML element is displayed and how space is allocated around it. Every element on a webpage is treated as a rectangular box consisting of four components

- content: The actual text, image, or other content inside the element.
- padding: Space between the content and the border, making the content easier to read.
- Border: A line that surrounds the padding and content.
- Margin: The outermost space that separates an element from other elements.

Example:-

```
• box {  
  width : 200px;  
  padding : 20px;  
  border : 2px solid black;  
  margin : 15px;  
}
```

CSS positioning Techniques:-

CSS positioning controls how elements are placed on a webpage

- **Static**: Default positioning based on the normal document flow
- **Relative**: Moves an element relative to its original position.
- **Absolute**: positions an element relative to its nearest positioned parent
- **Fixed**: keeps an element fixed relative to the browser window even when scrolling.
- **Sticky**: Behaves like a relative element until a specific scroll position, then becomes fixed

Proper use of positioning ensures a well-structured and organized webpage layout

using media Queries to Resolve layout issues:-

Media Queries apply different CSS styles based on the screen size, allowing the layout to adapt to desktops, tablets, and mobile devices.

Example:-

```
• container {  
  display: flex;  
}  
@media (max-width: 768px) {  
  • container {  
    flex-direction: column;  
  }  
}
```

Best practices for a Suitable and Responsive layout:-

- use flexbox or CSS grid for flexible layouts
- use percentage (%), rem, em or vw units instead of fixed pixels.
- Add media queries for different screen sizes.
- make images responsive using:

```
img {  
  max-width : 100%;  
  height : auto;  
}
```

- use the viewport meta tag:

```
<meta name="viewport"  
  Content="width=device-width,  
  initial-scale=1.0">
```

- Test the website on multiple browsers and devices
- Keep the layout simple and maintain consistent spacing

Conclusion:-

The CSS Box model defines the structure of every web page element through content, padding, border, and margin. CSS positioning determines how elements are arranged on the page while media queries ensures that layouts adapt smoothly to different screen sizes. By using Responsive design techniques, developers can create suitable attractive, and user-friendly websites that work effectively on desktops, tablets, and mobile devices.

(4)

Document object Model (DOM)

The Document object model (DOM) is a programming interface for HTML and XML documents. It represents a webpage as a tree-like structure of objects, where every HTML element, attribute, and text is treated as a node. JavaScript uses the DOM to access, modify, add or remove elements dynamically without reloading the webpage.

Role of the DOM in web development

- Allows JavaScript to access and modify HTML elements
- Enables dynamic updates to webpage content
- Changes CSS styles dynamically.
- Handles user interactions such as clicks, keyboard input, and mouse events
- Supports creating interactive and responsive web app

JavaScript DOM manipulation

- `getElementById()`
- `querySelector()`
- `getElementByClassName()`
- `createElement()`
- `appendChild()`
- `removeChild()`
- `innerHTML`
- `textContent`

These methods allow developers to update webpage content instantly without refreshing the page.

Example:-

```
<! Document-type html >
<html>
<head>
<title> Dom Example </title>
</head>
<h2 id="message">welcome! </h2>
<button onclick="change text()">click here </button>
<script>
function change text () {
    document.getElementById ("message").innerHTML =
        "welcome to internet programming" ;
}
</script>
</body>
</html>
```

Improvements for Better performance and user experience

- use efficient DOM selection methods like query selector () when appropriate.
- minimize unnecessary DOM updates to improve performance.
- use event listeners (add event listener()). Instead of inline event handler.

Conclusion:-

The Document object model (DOM) enables Javascript to interact with HTML elements dynamically, making webpages interactive and responsive. By manipulating the DOM, developers can update content, change styles, and respond to user actions without reloading the page. using efficient DOM manipulation techniques.